

## EDUCATION

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| <b>Bachelor of Engineering</b>   <i>Communication Engineering (Joint Program)</i>                                                                                                                                                                                                                                     | Sep. 2022 – Jul. 2026 |
| University of Electronic Science and Technology of China (UESTC) & University of Glasgow                                                                                                                                                                                                                              | Chengdu, China        |
| <ul style="list-style-type: none"><li>GPA: 84.9/ 3.73</li><li><b>Core Courses:</b><ul style="list-style-type: none"><li>* Linear Algebra and Space Analytic Geometry: 98</li><li>* Elements of Information Theory: 88</li><li>* Introductory Programming: 95</li><li>* Digital Circuit Design: 92</li></ul></li></ul> |                       |

## HONORS AND AWARDS

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|--------------------------------------------------------------------|-----------|
| <b>Outstanding student of the university</b>                       | Oct. 2024 |
| Awarded students who are outstanding both academically and morally |           |
| <b>Outstanding student of the university</b>                       | Oct. 2023 |
| Awarded students who are outstanding both academically and morally |           |
| <b>Outstanding students of the college</b>                         | Oct. 2023 |
| Recognition for to 15% of students in academics at the college     |           |

## PUBLICATIONS

### Conference paper

- **J. Guo**, L. Zhang, and Y. Shen, "A Human-Centered Multimodal AutoML Framework for Discriminative and Generative Tasks with Large Language Models," in IEEE BIBM 2025.

## RESEARCH EXPERIENCE

### Diffusion Models for Virtual Try-On Systems

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| <b>National Supercomputing Center</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Sep. 2024 – Apr. 2025 |
| Supervisor: Dr. Changwei Wang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Jinan, China          |
| <ul style="list-style-type: none"><li>• Developed a virtual try-on framework applying diffusion models and other advanced techniques, enabling users to easily visualize clothing changes using just a single image.</li><li>• Enhanced the realism and accuracy of garment fitting, improving user experience and convenience in online shopping scenarios</li><li>• Addressed the challenge of replacing long-sleeved clothes with short-sleeved garment by staged and partial replacment, largely improving user experience.</li></ul> |                       |

### Generative AI in Semantic Communication Systems

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| <b>UESTC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Apr. 2024 – May. 2025 |
| Supervisor: Prof. Yusha Liu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Chengdu, China        |
| <ul style="list-style-type: none"><li>• Applied generative AI model to suppress information to its latent space and decompress after transmission which conserved communication resources in communication systems</li><li>• Explored the method to transmit information accurately under water using AI models to fix inaccurate received information by learning large numbers of underwater communication datasets</li><li>• Solved the problem of applying AI models on complex-valued communication systems and modify my AI model since AI models are mostly used for real-valued problems while information transmission often uses real values</li></ul> |                       |

## Large Language Models for Automated Machine Learning

**Johns Hopkins University**

Supervisor: Dr. Yiqing Shen

Feb. 2024 – May. 2025

Baltimore, MD

- Led a research project independently on applying large language models on conducting both discriminative and generative machine learning automatically
- Conducted experiments on multiple datasets in different conditions using my framework, which performed better in most situations
- Performed a user experiment with 25 participants and the results validated its convenience
- Wrote a research paper independently based on the project and accepted by IEEE BIBM 2025

## PROJECT EXPERIENCES

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### Automated Pace Tracking Drone

**UESTC**

Jan. 2025 - Jul. 2025

Chengdu, China

- Designed and constructed a drone that can achieve automatic tracking, recognize objects and pick them up
- Constructed a drone from the very beginning including programming, building the fuselage, verifying its capabilities without a Supervisor
- Completely autonomous without using remote sensing, the drone trace path, recognize and grasp objects by itself

### Mini Model of Smart Home Using Embedded Processor

**UESTC**

Apr. 2023 - Jun. 2023

Chengdu, China

- Constructed a mini model of smart home using STM32 embedded processor, which can detect environmental factors like temperature, luminance, humidity and respond accordingly
- Independently program the embedded processor, constructed the circuit and tested it

### Simple AI model for classifying objects

**UESTC**

Dec. 2024 - Jan. 2025

Chengdu, China

- Constructed a AI model using Transformer to classify 7 types of objects
- Contributed to the dataset we used and my model performed better than other students' models

## EXTRACURRICULAR ACTIVITIES

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**Reviewer of CVPR**

Dec. 2024 - Jan. 2025

Worked as a reviewer, reviewing papers of medical computer vision frameworks

**Champion of the College Football Tournament**

Oct. 2022

Recognized as the Champion Class in the Tournament

**Outstanding Class Committee Member of the College**

Oct. 2023, Oct. 2024

Granted to class committee members who demonstrated exceptional dedication

## SKILLS

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**Languages:** English(IELTS: 7.5)

**Programming:** Python (NumPy, SciPy, Matplotlib, Pandas, Pytorch), MATLAB, C

**Document Creation:** Microsoft Office Suite, LaTeX